

Name: Adalynn Cadence Everly

ID: 1001852753

Class: CSE-4321-001

HW 3:

(No instructions)

CSE 4321: Software Testing and Maintenance

HW 3 (30 points)

Consider a system that consists of 4 parameters, P1, P2, P3, and P4. Each parameter has two values 0 and 1. Apply algorithm IPO to create a pairwise test set for this system. Use “-” to represent *don't care* values, i.e., values that do not affect coverage. Clearly indicate your tie-breaking rules that may be needed in the test generation process. You must show intermediate steps to obtain full credits.

4 parameters P_1, P_2, P_3 & P_4 2 val = 0, 1. IP6 use -

To represent don't care Vals

so, $P_1 \rightarrow P_2: (0,0), (0,1), (1,0), (1,1)$

$P_1 \rightarrow P_3, P_1 \rightarrow P_4, P_2 \rightarrow P_3, P_2 \rightarrow P_4, P_3 \rightarrow P_4$ so... $C(4,2) \times 4$
 $6 \times 4 = 24$ vals that are different.

Double Check:

so init table

P_1	P_2
0	0
0	1
1	0
1	1

$P_1 \rightarrow P_2$
 Now let's do P_3 (horizontal)

P_1	P_2	P_3	pairs
0	0	0	$(0,0), (0,0)$ P_1, P_3
0	1	1	$(0,1), (1,1)$ P_2, P_3
1	0	1	$(1,0), (0,1)$
1	1	0	$(1,0), (1,0)$

$+ P_4$ (Ver + + Row insert)

P_1	P_2	P_3	P_4	pairs
0	0	0	0	$(0,0) \times 3$
0	1	1	1	$(0,1), (1,1), (1,1)$
1	0	1	0	$(1,0), (0,0), (1,0)$
1	1	0	1	$(1,1), (1,1), (0,1)$

Fin Test Answer All

P_1	P_2	P_3	P_4
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	1

so... all $P(1,4), (2,4), (3,4)$
 have: $(0,0), (0,1), (1,0), (1,1)$

Answer:

P_1	P_2	P_3	P_4
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	1
0	-	0	1
1	-	1	1

Comment?

Tie Breaking rules

when + ing a new parameter, I prioritized existing rows based on the max net pair gain. Not just what new pairs were covered, but also if they overapped w/ earlier 1's in an efficient way. When there were ties, I picked the val. that minimized the entropy across the row. so I kept the ~~less~~ more clustered b/c that often led to fewer new rows in the long run. If that still didn't break the tie, I defaulted to keeping the row lexicographically balanced just to keep the pattern clear for debugging. ~~for later~~
 so I optimized coverage 1st, chaos 2nd.